CORNELL UNIVERSITY
Mechanical and Aerospace Engineering

M&AE 6760: Model Based Estimation

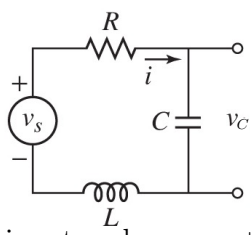
Prof. M. Campbell | **Homework #1** | Date: 22 Jan 2025
Spring 2026 | Due: (5pm) 6 Feb 2025

Note that suggested MATLAB *.m commands are given in some problems. These are only suggestions, as there may be other commands or Python equivalents you could use.

Problem #1: State Space Modeling

Please complete the following state space modeling problems:

a) Find a state space model of the following 'series' RLC circuit (coupled ODEs given to the right):



$$L\frac{di}{dt} = v_S - v_C - iR$$
$$C\frac{dv_C}{dt} = i$$

where $u = v_S$ as the input and $y = v_C + n$ as the output, where n is the sensor noise. In this part, please use a state vector:

$$\mathbf{x} = \begin{bmatrix} i \\ v_C \end{bmatrix}$$

b) Find a transformation T such that a new state vector is formed from that in (a):

$$\tilde{\mathbf{x}} = \begin{bmatrix} v_C \\ \frac{dv_C}{dt} \end{bmatrix} = T\mathbf{x}$$

Outline how you would find the state space model using this state vector using T.

c) Assuming $R = 5\Omega$, $C = 0.25$F and $L = 1$H (all SI units, no conversions required), find the eigenvalues and eigenvectors for your state space system in (a). The MATLAB function eig.m may be used. Explain in your own words how these can be used to solve for the 'free' response.

d) Using the state space model from (c), check the observability of the system. The MATLAB function obsv.m may be used.

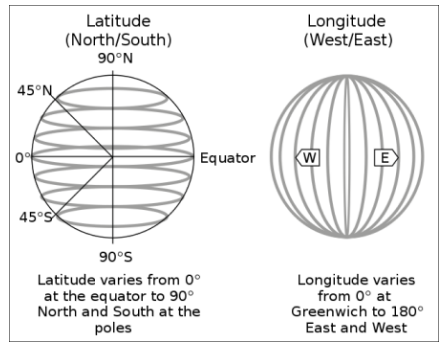
e) Create a sensor model $y_{\min} = C\mathbf{x}$ that only measures the system response associated with the smallest eigenvalue, $|\lambda_{\min}|$? Explain in your own works if this this a good sensor (or not).

Problem #2: **Multivariate Gaussian Random Variable**

You are sitting in your MAE 6760 class, somewhat bored with the probability lectures - when you check your phone. You want to know if your phone can localize your position accurately indoors. So, you run a few tests.

First, you note that the true Latitude and Longitude of your location (using Upson Hall in Google Maps) is

$$\mathbf{x}_{\mathsf{true}} = \begin{bmatrix} 42.44396 \text{ deg} \\ -76.48248 \text{ deg} \end{bmatrix}$$



Then, you hack into your phone and pull out the solution to the localization filter. The localization estimator running on your phone produces an estimate of your location in 'LLA' coordinates: Latitude, Longitude, and Altitude as a Gaussian distribution. Currently, your phone estimate is:

$$\mathbf{x}_{\mathsf{LLA}} = \begin{bmatrix} \text{Latitude (in deg)} \\ \text{Longitude (in deg)} \\ \text{Altitude (in m above sea level)} \end{bmatrix} \qquad \longrightarrow \qquad \mathbf{x}_{\mathsf{LLA}} \sim \mathcal{N}(\hat{\mathbf{x}}_{\mathsf{LLA}}, P_{\mathsf{LLA}})$$

where the mean $\hat{\mathbf{x}}_{\mathsf{LLA}}$ and covariance P_{LLA} are given as

$$\hat{\mathbf{x}}_{\mathsf{LLA}} = \begin{bmatrix} 42.44445 \text{ deg} \\ -76.48252 \text{ deg} \\ 263 \text{ m} \end{bmatrix}, \qquad P_{\mathsf{LLA}} = \begin{bmatrix} 400 & 40 & 100 \\ 40 & 400 & 100 \\ 100 & 100 & 2500 \end{bmatrix} \quad \text{all in m}^2$$

where the covariance has been converted to units of m^2 (as described in (d)). Please complete the following:

a) First, find the Latitude and Longitude only distribution by marginalizing out the Altitude of the 3D distribution. An alternative is to use a transformation $\mathbf{x}_{\mathsf{LL}} = C\mathbf{x}_{\mathsf{LLA}}$; what would C be in this case?

b) Sketch the '1-σ' or 68% contour for the 2D Gaussian distribution for $\mathbf{x}_{\mathsf{LL}}$ from (a) around the [0;0] error (using the ellipsoid MATLAB script on Canvas or by hand). Note the ellipsoid dimensions (max/min, East/North) and discuss how accurate this is (compare to the size of a classroom or building or other).

c) We want to develop a simple metric to evaluate if the estimator is accurate. You decide that the estimator **IS** accurate if the true location falls within the 95% uncertainty of the estimator (equivalent to the '2-σ' probability contour of $\mathbf{x}_{\mathsf{LL}}$). The '1-σ' probability contour of $\mathbf{x}_{\mathsf{LL}}$ is:

$$\mathbf{e}_{\mathsf{m}}^T P_{\mathsf{LL}}^{-1} \mathbf{e}_{\mathsf{m}} = [\mathbf{x}_{\mathsf{LL}} - \hat{\mathbf{x}}_{\mathsf{LL}}]^T P_{\mathsf{LL}}^{-1} [\mathbf{x}_{\mathsf{LL}} - \hat{\mathbf{x}}_{\mathsf{LL}}] = 1$$

where the error and covariance are in latitude/longitude. You are to derive a test as to whether the error is acceptable or not (i.e. the error is within the '2-σ' probability contour of $\mathbf{x}_{\mathsf{LL}}$) of the form:

$$f(\mathbf{e}_{\mathsf{m}}, P_{\mathsf{LL}}) < 1$$

i.e. if $f < 1$, then the error is within the 95% level (and the estimator is working well); if $f > 1$, then the error is outside the 95% level (and the estimator is not working well); and $f = 1$ in the border, which is the 95% probability contour ellipse.

d) Finally, calculate the error of your estimator using

$$\mathbf{e}_{\mathsf{deg}} = \mathbf{x}_{\mathsf{true}} - \hat{\mathbf{x}}_{\mathsf{LL}}$$

Convert this error to meters by approximating the Earth as a sphere and considering small angles,

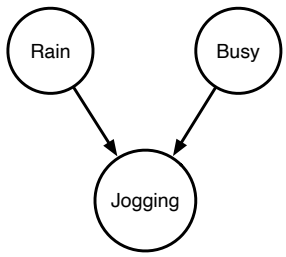
$$\mathbf{e}_{\mathsf{m}} = \mathbf{e}_{\mathsf{deg}} \frac{\pi \text{ rad}}{180 \text{ deg}} \times 6378100 \text{ m}$$

Is the estimator working well?

Problem #3: Bayesian Probability and Inference

Prof. Campbell likes to jog, and has two key factors driving his decision-making each day: whether it is raining or not, and how busy he is. Below is a probabilistic graphical model which captures these two factors in his decision-making. The model includes prior probabilities of rain ($R \in \{\text{yes,no}\}$), and how busy he is ($B \in \{\text{low,med,high}\}$). A conditional probability table is given for his decision-making about whether to jog $P(J|R, B)$, where $J \in \{\text{yes,no}\}$.

$P(R = \text{yes})$	$P(R = \text{no})$
0.3	0.7



$P(B = \text{low})$	$P(B = \text{med})$	$P(B = \text{high})$
0.2	0.5	0.3

R	B	$P(J = \text{yes}\|R, B)$	$P(J = \text{no}\|R, B)$
yes	low	0.4	0.6
yes	med	0.2	0.8
yes	high	0.1	0.9
no	low	0.9	0.1
no	med	0.7	0.3
no	high	0.5	0.5

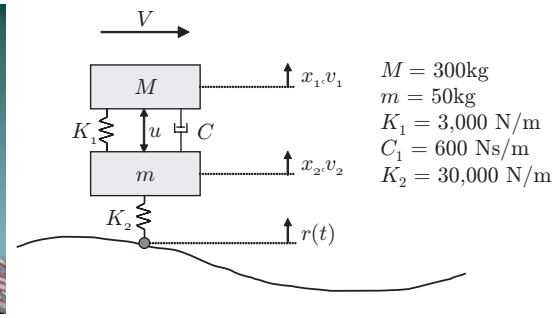
a) Calculate the probability that Prof. Campbell jogs on any given day, $P(J = \text{yes})$.

b) Evaluate the key factors in Prof. Campbell's decision making, by calculating the probability of his decision being driven more by rain $P(J = \text{yes}|R = \text{no})$ or how busy he is $P(J = \text{yes}|B = \text{low})$.

c) If you see Prof. Campbell jogging, what is the probability he was very busy that day, $P(B = \text{high}|J = \text{yes})$?

Problem #4: Linear System driven by Random Process

We are going to evaluate the performance of an open loop and closed loop suspension system. Consider the simplified model of car below:



The model includes the following variables:

x_1, v_1	position,velocity of car/driver	x_2, v_2	position,velocity of wheels/tires
M	mass of the car/driver	m	mass of wheels, axles, etc.
K_1	spring of passive shocks	K_2	stiffness of the tires
C_1	damper of passive shocks		
$u(t)$	hydraulic actuator	$r(t)$	roadway height w.r.t an inertial frame

The problem m-file (from Canvas) gives the state space model for this system, which includes the road $r(t)$ and actuator $u(t)$ as inputs, and the performance output $z(t) = x_1(t)$ as the position of the driver. The road can be modeled as continuous time white noise with intensity $\Sigma_r = 10^{-4}\text{m}^2/\text{sec}$. Note that the positions/velocities are measured from the unsprung condition (i.e. 0).

a) Simulate the 'open loop' system for the car driving over a bumpy road via lsim.m ($r \to z$). Calculate the sample based variance of the performance z via cov.m, and compare to the analytical solution via the continuous time Lyapunov equation lyap.m (e.g. percent difference or some other metric). What is the key reason these are different?

b) The m-file also includes the solution for a state feedback controller $u = -K\mathbf{x}$ via LQR control. The goal of the controller is to improve (reduce) the motion of driver ($z = x_1$). Using the open loop state space model and the control law, find the closed loop state space model by hand. Using this model, repeat the analysis from (a), i.e. simulate the system and compare sample vs analytical performance (variance of z).

c) Plot the performance z for the open and closed loop system over time (from (a) and (b)). How much does the controller improve the standard deviation of the performance, σ_z? For each case, what is the probability that the driver will move more than 3cm (from the unsprung caase) at any given time? You can use normcdf.m.

d) Calculate the variance of the control input $u(t)$ analytically (via lyap.m), and via samples (cov.m or var.m) and compare the results. Note: you should see that this case has a larger difference (%) than (a) and (b). Any ideas for why this is? Note 2: this is hard :).